

MA 378 Problems 2

Please complete the questions marked with (★) as a written assignment. Submit a pdf of your answers in the Assessment section of Blackboard. The deadline for submission will be 11pm, Friday 7 May.

1. Let $q_0, q_1, \dots, q_n$ be the quadrature weights for the Newton-Cotes rule $Q_n(f)$, used to estimate $\int_a^b f(x)dx$.

(a) Show that $q_i = q_{n-i}$ for $i = 0, \dots, n$.

(b) Show that $\sum_{i=0}^n q_i = b - a$.

2. Deduce the 4-point Newton-Cotes Rule for estimating the integral $\int_0^1 f(x)dx$:

$$Q_3(f) = q_0 f(x_0) + q_1 f(x_1) + q_2 f(x_2) + q_3 f(x_3).$$

Extend the rule to estimate the integral of functions over $[a, b]$.

3. Determine the precision of the following schemes for estimating $\int_0^1 f(x)dx$.

(a) $Q(f) = f(\frac{1}{2})$.

(b) $Q(f) = \frac{1}{4}f(0) + \frac{3}{4}f(\frac{2}{3})$.

(c) (★) $Q(f) = \frac{3}{2}f(\frac{1}{3}) - 2f(\frac{1}{2}) + \frac{3}{2}f(\frac{2}{3})$.

4. (★) Show that it is impossible to choose $n + 1$ quadrature points and weights so that the $n + 1$ -point quadrature rule

$$\int_a^b f(x)dx \approx \sum_{k=0}^n q_k f(x_k)$$

has precision $2n + 2$. (*Hint: To show the method does not have precision $2n + 2$, you just need to give an example of a single polynomial p of degree exactly $2n + 2$ for which $\int_a^b p(x)dx \neq \sum_{k=0}^n q_k f(x_k)$.)*

5. We saw in lectures that the Gaussian quadrature rule

$$\int_{-1}^1 f(x)dx \approx G_1(f) := f\left(\frac{-1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$$

is exact for all polynomials of degree 3 or less.

(a) Use a change of variables to extend the $G_1(f)$ rule to an arbitrary interval $[a, b]$.

(b) Use this rule to estimate $\int_1^2 \ln(x)dx$.

6. (★) Consider the differential equation

$$-u''(x) + u(x) = 3x \quad \text{on} \quad (-1, 2)$$

with $u(-1) = u(2) = 0$. State the weak/variational formulation of this differential equation.

7. (★) Suppose one wanted to compute an approximation to the differential equation of the previous question using the finite element method with the usual piecewise linear Galerkin basis functions on the uniform mesh $\{x_0, x_1, x_2, x_3\} = \{-1, 0, 1, 2\}$. Give a formula for the basis functions. How is the FEM solution expressed in terms of this basis?

MA 378 Problems 1

Please complete the questions marked with (★) as a written assignment. Submit a pdf of your answers in the Assessment section of Blackboard.

1. Show that $\sum_{i=0}^n L_i(x) = 1$ for all x , where $L_i(x)$ is the Lagrange interpolant.
2. Write down the Lagrange Form of p_2 the polynomial of degree 2 that interpolates the points $(0, 3)$, $(1, 2)$ and $(2, 4)$.
3. (★) Let p_3 be the polynomial that interpolates $f(x) = \sin(x/2)$ at the points $x_0 = 0$; $x_1 = 1/3$; $x_2 = 2/3$ and $x_3 = 1$. Use Cauchy's theorem to give a bound for the error at $x = 1/2$ (You don't have to give a formula for p_3).
4. (★) Given the distinct points (x_i, y_i) , $i = 0, 1, \dots, n+1$, let q be the Lagrange polynomial of degree n for the set of points $\{(x_i, y_i) : i = 0, 1, \dots, n\}$ and let r be the Lagrange polynomial of degree n for the points $\{(x_i, y_i) : i = 1, 2, \dots, n+1\}$. Define

$$p(x) = \frac{(x - x_0)r(x) - (x - x_{n+1})q(x)}{x_{n+1} - x_0}.$$

Show that p is the Lagrange polynomial of degree $n + 1$ for the points $\{(x_i, y_i) : i = 0, 1, \dots, n + 1\}$.

5. (★) Verify that

$$p_{2n+1}(x) = \sum_{i=0}^n (f(x_i)H_i(x) + f'(x_i)K_i(x))$$

is the solution to the Hermite Polynomial Interpolation problem.

6. (★) Construct the Hermite interpolation polynomial of degree 3 for the function $f(x) = x^5$, using the points $x_0 = 0$ and $x_1 = a$, for some $a \in \mathbb{R}$, and show that it has the form $p_3(x) = 3a^2x^3 - 2a^3x^2$.
7. Suppose we used a cubic polynomial to interpolate four data points, each of which give the maximum temperatures recorded on one of four consecutive days. Briefly explain in your own words why it would be a bad idea to make predictions about the temperature in the future using this model. (You may remember reports from a country that will remain nameless, that suggested at some point around March 2020 that COVID cases would hit zero in May or June "based on their cubic model". You might similarly see why this was unrealistic, especially if you consider what that model would say if used to predict cases from July onwards.)

Assignment 2

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3.

$$(c) Q(f) = \frac{3}{2}f\left(\frac{1}{3}\right) - 2f\left(\frac{1}{2}\right) + \frac{3}{2}f\left(\frac{2}{3}\right)$$

$$f(x) = 1 \quad Q(f) = \frac{3}{2} - 2 + \frac{3}{2} = 1$$

$$\int_0^1 dx = 1$$

$$f(x) = x \quad Q(f) = \left(\frac{3}{2}\right)\left(\frac{1}{3}\right) - 2\left(\frac{1}{2}\right) + \frac{3}{2}\left(\frac{2}{3}\right) = \frac{1}{2}$$

$$\int_0^1 x dx = \frac{1}{2}$$

$$f(x) = x^2 \quad Q(f) = \left(\frac{3}{2}\right)\left(\frac{1}{9}\right) - 2\left(\frac{1}{4}\right) + \frac{3}{2}\left(\frac{4}{9}\right) = \frac{1}{3}$$

$$\int_0^1 x^2 dx = \frac{1}{3}$$

$$f(x) = x^3 \quad Q(f) = \left(\frac{3}{2}\right)\left(\frac{1}{27}\right) - 2\left(\frac{1}{8}\right) + \frac{3}{2}\left(\frac{8}{27}\right) = \frac{1}{4}$$

$$\int_0^1 x^3 dx = \frac{1}{4}$$

$$f(x) = x^4 \quad Q(f) = \frac{3}{2}\left(\frac{1}{64}\right) - 2\left(\frac{1}{16}\right) + \frac{3}{2}\left(\frac{16}{64}\right) \neq \frac{1}{5}$$

$$\int_0^1 x^4 dx = \frac{1}{5}$$

Precision $n = 4$

4.

Let π_{n+1} be the nodal polynomial associated with the quadrature points:

$$\pi_{n+1}(x) = (x - x_0)(x - x_1) \dots (x - x_n)$$

$$\text{Let } f(x) = (\pi_{n+1}(x))^2 \in P_{2n+2},$$

note that $f(x) \geq 0$ for all x , so $\int_a^b f(x) dx > 0$

but $f(x_i) = 0$ for all i so

$$I_n(f) = \sum_{k=0}^n w_k f(x_k) = 0$$

so I_n is not exact for this polynomial of degree $2n+2$
therefore doesn't have precision $2n+2$

6. $-u''(x) + u(x) = 3x$ on $(-1, 2)$ with $u(-1) = u(2) = 0$

The weak/variational formulation:

Find $u \in H_0^1(a, b)$ such that:

$$A(u, v) = L(v) \quad \text{for all } v \in H_0^1(a, b)$$

where $A(u, v) := (u', v') + (ru, v)$ and inner product $(u, v) := \int_a^b u(x)v(x)dx$
 $L(v) = (f, v)$

$$\int_a^b u'(x)v'(x)dx + \int_a^b r(x)u(x)v(x)dx = \int_a^b f(x)v(x)dx$$

Here, $b=2$, $a=-1$, $r(x)=1$, $f(x)=3x$:

Find $u \in H_0^1(-1, 2)$ such that

$$\int_{-1}^2 u'(x)v'(x)dx + \int_{-1}^2 u(x)v(x)dx = \int_{-1}^2 3xv(x)dx \quad \text{for all } v \in H_0^1(-1, 2)$$

7. $-u''(x) + u(x) = 3x$ on $(-1, 2)$ with $u(-1) = u(2) = 0$

mesh: $\{x_0, x_1, x_2, x_3\} = \{-1, 0, 1, 2\}$

The FEM is:

Find $u_h(x) \in S$ such that

$$a(u_h, v_h) := (u_h', v_h') + (u_h, v_h) = 3(x, v_h) \text{ for all } v_h(x) \in S$$

$h = \frac{2 - (-1)}{3} = 1$, so let

$$\varphi_1(x) = \begin{cases} x+1 & -1 < x < 0 \\ 1-x & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\varphi_2(x) = \begin{cases} x & 0 < x < 1 \\ 2-x & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

and $u_h(x) = \lambda_1 \varphi_1(x) + \lambda_2 \varphi_2(x)$

$$(\lambda_1 \varphi_1' + \lambda_2 \varphi_2', \varphi_1') + (\lambda_1 \varphi_1 + \lambda_2 \varphi_2, \varphi_1) = 3(x, \varphi_1)$$

$$(\lambda_1 \varphi_1' + \lambda_2 \varphi_2', \varphi_2') + (\lambda_1 \varphi_1 + \lambda_2 \varphi_2, \varphi_2) = 3(x, \varphi_2)$$

$$\textcircled{1} \lambda_1 \left(\int_{-1}^2 \varphi_1' \varphi_1' dx + \int_{-1}^2 \varphi_1 \varphi_1 dx \right) + \lambda_2 \left(\int_{-1}^2 \varphi_2' \varphi_1' dx + \int_{-1}^2 \varphi_2 \varphi_1 dx \right) = 3 \int_{-1}^2 x \varphi_1 dx$$

$$\textcircled{2} \lambda_1 \left(\int_{-1}^2 \varphi_1' \varphi_2' dx + \int_{-1}^2 \varphi_1 \varphi_2 dx \right) + \lambda_2 \left(\int_{-1}^2 \varphi_2' \varphi_2' dx + \int_{-1}^2 \varphi_2 \varphi_2 dx \right) = 3 \int_{-1}^2 x \varphi_2 dx$$

$$\textcircled{1} \int_{-1}^2 \varphi_1' \varphi_1' dx = \int_{-1}^0 (1)^2 dx + \int_0^1 (-1)^2 dx = \left. x \right|_{-1}^0 + \left. x \right|_0^1 = -(-1) + 1 = 2$$

$$\int_{-1}^2 \varphi_1 \varphi_1 dx = \int_{-1}^0 (x+1)^2 dx + \int_0^1 (1-x)^2 dx = \left. \frac{x^3}{3} + x^2 + x \right|_{-1}^0 + \left. x - x^2 + \frac{x^3}{3} \right|_0^1 = -\left(-\frac{1}{3} + 1 - 1\right) + \left(1 - 1 + \frac{1}{3}\right) = \frac{2}{3}$$

Coefficient of λ_1 is $\frac{8}{3}$

$$\int_{-1}^2 \psi_2' \psi_1' dx = \int_0^1 (1)(-1) dx = -x \Big|_0^1 = -1$$

$$\int_{-1}^2 \psi_2 \psi_1 dx = \int_0^1 (x)(1-x) dx = \frac{x^2}{2} - \frac{x^3}{3} \Big|_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

Coefficient of λ_2 is $-1 + 1/6 = -5/6$

$$\begin{aligned} 3 \int_{-1}^2 x \psi_1 dx &= 3 \left(\int_{-1}^0 x(x+1) dx + \int_0^1 x(1-x) dx \right) = 3 \left(\frac{x^3}{3} + \frac{x^2}{2} \Big|_{-1}^0 + \frac{x^2}{2} - \frac{x^3}{3} \Big|_0^1 \right) \\ &= 3 \left(-\left(\frac{1}{3} + \frac{1}{2}\right) + \frac{1}{2} - \frac{1}{3} \right) \\ &= 0 \end{aligned}$$

$$\textcircled{1} \quad 16\lambda_1 - 5\lambda_2 = 0$$

②

$$\int_{-1}^2 \psi_1' \psi_2' dx = -1$$

$$\int_{-1}^2 \psi_1 \psi_2 dx = \frac{1}{6}$$

Coefficient of λ_1 is $-1 + 1/6 = -5/6$

$$\int_{-1}^2 \psi_2' \psi_2' dx = \int_0^1 (1)^2 dx + \int_1^2 (-1)^2 dx = x \Big|_0^1 + x \Big|_1^2 = 1 + (2-1) = 2$$

$$\int_{-1}^2 \psi_2 \psi_2 dx = \int_0^1 x^2 dx + \int_1^2 (2-x)^2 dx = \frac{x^3}{3} \Big|_0^1 + \frac{4x - 2x^2 + \frac{x^3}{3}}{1} \Big|_1^2 = \frac{1}{3} + \left(8 - 8 + \frac{8}{3} - \left(4 - 2 + \frac{1}{3} \right) \right) = \frac{2}{3}$$

Coefficient of λ_2 is $2 + 2/3 = 8/3$

$$\begin{aligned} 3 \int_{-1}^2 x \psi_2 dx &= 3 \left(\int_0^1 x^2 dx + \int_1^2 x(2-x) dx \right) = 3 \left(\frac{x^3}{3} \Big|_0^1 + x^2 - \frac{x^3}{3} \Big|_1^2 \right) = 3 \left(\frac{1}{3} + \left(4 - \frac{8}{3} - \left(1 - \frac{1}{3} \right) \right) \right) \\ &= 3(1) \\ &= 3 \end{aligned}$$

$$\textcircled{2} \quad -5\lambda_1 + 16\lambda_2 = 18$$

$$16\lambda_1 - 5\lambda_2 = 0$$

$$-5\lambda_1 + 16\lambda_2 = 18$$

$$\lambda_1 = \frac{5}{16} \lambda_2$$

$$-5\left(\frac{5}{16}\lambda_2\right) + 16\lambda_2 = 18$$

$$-25\lambda_2 + 256\lambda_2 = 288$$

$$231\lambda_2 = 288$$

$$\lambda_2 = \frac{96}{77}$$

$$\lambda_1 = \frac{5}{16} \left(\frac{96}{77}\right)$$

$$= \frac{30}{77}$$

$$u_n = \lambda_1 \varphi_1(x) + \lambda_2 \varphi_2(x) = \frac{30}{77} \varphi_1(x) + \frac{96}{77} \varphi_2(x)$$

MA378 : Problem Sheet 1

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Q.1 Show that $\sum_{i=0}^n L_i(x) = 1$ for all x .

Interpolating
Polynomial :

$$p(x) = \sum_{i=0}^n f(x_i) L_i(x)$$

for $f(x) = 1$

$$p(x) = \sum_{i=0}^n L_i(x)$$

For all x , $p(x) - f(x) = 1$

$$\therefore \sum_{i=0}^n L_i(x) = 1$$

Q.2 $x_0 = (0, 3)$, $x_1 = (1, 2)$, $x_2 = (2, 4)$

Find Lagrange form of p_2

$$L_0(x) = \frac{(x-1)(x-2)}{(0-1)(0-2)}$$

$$L_1(x) = \frac{(x-0)(x-2)}{(1-0)(1-2)}$$

$$L_2(x) = \frac{(x-0)(x-1)}{(2-0)(2-1)}$$

$$p_2(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x)$$

$$\frac{3(x^2-3x+2)}{2} + \frac{2(x^2-2x)}{-1} + \frac{4(x^2-x)}{2}$$

$$\frac{3x^2}{2} - \frac{9x}{2} + 3 - 2x^2 + 4x + 2x^2 - 2x$$

$$p_2(x) = \frac{3}{2}x^2 - \frac{5}{2}x + 3$$

Q3. $f(x) = \sin\left(\frac{x}{2}\right)$, $x_0 = 0$, $x_1 = \frac{1}{3}$, $x_2 = \frac{2}{3}$, $x_3 = 1$

Cauchy's Theorem: $f(x) - p_n(x) = \frac{f^{(n+1)}(\tau)}{(n+1)!} \pi_{n+1}(x)$

$$f(x) = \sin\left(\frac{x}{2}\right)$$

$$f'(x) = \frac{1}{2} \cos\left(\frac{x}{2}\right)$$

$$f''(x) = -\frac{1}{4} \sin\left(\frac{x}{2}\right)$$

$$f'''(x) = -\frac{1}{8} \cos\left(\frac{x}{2}\right)$$

$$f^{(4)}(x) = \frac{1}{16} \sin\left(\frac{x}{2}\right)$$

$$\begin{aligned} \pi_4(x) &= (x-0)(x-\frac{1}{3})(x-\frac{2}{3})(x-1) \\ &= x^4 - 2x^3 + \frac{11x^2}{9} - \frac{2x}{9} \end{aligned}$$

$$f(x) - p_3(x) = \frac{f^{(4)}(\tau)}{4!} \pi_4(x)$$

$$f(x) - p_3(x) = \frac{\frac{1}{16} \sin\left(\frac{\tau}{2}\right)}{24} \left(x^4 - 2x^3 + \frac{11x^2}{9} - \frac{2x}{9}\right)$$

Choose $\tau \in [0,1]$ to maximise $\frac{1}{16} \sin\left(\frac{\tau}{2}\right)$
 $\sin(x)$ increases from $0 \rightarrow \frac{1}{2}$

$$\therefore \frac{1}{16} \sin\left(\frac{\tau}{2}\right) \leq \frac{1}{16} \sin\left(\frac{1}{2}\right) \text{ when } \tau \in [0,1]$$

$$|f(x) - p_3(x)| = \frac{\frac{1}{16} \sin\left(\frac{1}{2}\right)}{24} \left|x^4 - 2x^3 + \frac{11x^2}{9} - \frac{2x}{9}\right|$$

$$|f(x) - p_3(x)| = \frac{\sin\left(\frac{1}{2}\right)}{384} \left|\frac{9x^4 - 18x^3 + 11x^2 - 2x}{9}\right|$$

$$\begin{aligned} \left|f\left(\frac{1}{2}\right) - p_3\left(\frac{1}{2}\right)\right| &= \frac{\sin\left(\frac{1}{2}\right)}{384} \left|\frac{9\left(\frac{1}{2}\right)^4 - 18\left(\frac{1}{2}\right)^3 + 11\left(\frac{1}{2}\right)^2 - 2\left(\frac{1}{2}\right)}{9}\right| \\ &= \frac{\sin\left(\frac{1}{2}\right)}{384} \left(\frac{1}{144}\right) \end{aligned}$$

Bound for error of p_3 at $x = \frac{1}{2}$: $0.0000086702 = 8.67 \times 10^{-6}$

Q.4.

$q(x)$ is lagrange polynomial of degree n for $\{(x_i, y_i) : i=0, 1, \dots, n\}$

$r(x)$ is lagrange polynomial of degree n for $\{(x_i, y_i) : i=1, 2, \dots, n+1\}$

$$p(x) = \frac{\overbrace{(x-x_0)r(x)}^{\text{degree } n+1} - \overbrace{(x-x_{n+1})q(x)}^{\text{degree } n+1}}{x_{n+1}-x_0}$$

• $p(x)$ is of degree $n+1$

$$q(x_i) = y_i, \quad \forall i = 0, 1, \dots, n$$

$$r(x_i) = y_i, \quad \forall i = 1, 2, \dots, n+1$$

$$q(x_k) = r(x_k) = y_k \quad \text{for } 1 \leq k \leq n$$

$$\begin{aligned} \bullet p(x_k) &= \frac{(x_k - x_0)r(x_k) - (x_k - x_{n+1})q(x_k)}{x_{n+1} - x_0} \\ &= \frac{(x_k - x_0)y_k - (x_k - x_{n+1})y_k}{x_{n+1} - x_0} = \frac{(x_{n+1} - x_0)y_k}{x_{n+1} - x_0} = y_k \end{aligned}$$

$$\begin{aligned} \bullet p(x_0) &= \frac{(x_0 - x_0)r(x_0) - (x_0 - x_{n+1})q(x_0)}{x_{n+1} - x_0} \\ &= \frac{-(x_0 - x_{n+1})y_0}{x_{n+1} - x_0} \\ &= y_0 \end{aligned}$$

$$\begin{aligned} \bullet p(x_{n+1}) &= \frac{(x_{n+1} - x_0)r(x_{n+1}) - (x_{n+1} - x_{n+1})q(x_{n+1})}{x_{n+1} - x_0} \\ &= y_{n+1} \end{aligned}$$

$p(x)$ is of degree $n+1$ and interpolates the points $\{(x_i, y_i) : i=0, 1, \dots, n+1\}$
 $\therefore$ is the Lagrange polynomial

Q5. Verify that $p_{2n+1}(x) = \sum_{i=0}^n (f(x_i)H_i(x) + f'(x_i)K_i(x))$ solves the HPIP

Conditions to satisfy: degree of $2n+1$, $p_{2n+1} = f(x_i)$, $p'_{2n+1}(x_i) = f'(x_i)$

Recall $L_i(x_k) = \begin{cases} 1 & i=k \\ 0 & i \neq k \end{cases}$ for $i, k = 0, 1, \dots, n$

$L_i(x)$ is of degree n

$$H_i(x) = \underbrace{[L_i(x)]^2}_{2n} \underbrace{(1 - 2L_i'(x)(x-x_i))}_{+1}$$

• degree of $H_i(x)$ is $2n+1$

$$H_i'(x) = [L_i(x)]^2(-2L_i'(x)) + (1 - 2L_i'(x)(x-x_i)) \cdot 2L_i(x) \cdot L_i'(x)$$

if $i=k$

$$H_i(x_i) = [L_i(x_i)]^2 (1 - 2L_i'(x_i)(x_i - x_i))$$

$$H_i(x_i) = [1]^2 (1 - 0) = 1$$

$$\begin{aligned} H_i'(x_i) &= [L_i(x_i)]^2(-2L_i'(x_i)) + (1 - 2L_i'(x_i)(x_i - x_i)) \cdot 2L_i(x_i) \cdot L_i'(x_i) \\ &= 1^2(-2L_i'(x_i)) + (1 - 2L_i'(x_i)(0)) \cdot 2(1) \cdot L_i'(x_i) \\ &= -2L_i'(x_i) + 2L_i'(x_i) \\ &= 0 \end{aligned}$$

if $i \neq k$

$$\begin{aligned} H_i(x_k) &= [L_i(x_k)]^2 (1 - 2L_i'(x_k)(x_k - x_i)) \\ &= 0 \end{aligned}$$

$$\begin{aligned} H_i'(x_k) &= [L_i(x_k)]^2(-2L_i'(x_k)) + (1 - 2L_i'(x_k)(x_k - x_i)) \cdot 2L_i(x_k) \cdot L_i'(x_k) \\ &= 0 + (1 - 2L_i'(x_k)(x_k - x_i)) \cdot 0 \\ &= 0 \end{aligned}$$

$$H_i(x_k) = \begin{cases} 1 & i=k \\ 0 & i \neq k \end{cases}$$

$$H_i'(x_k) = 0$$

degree of $H(x)$ and $K(x)$ is $2n+1$

$\therefore \sum_{i=0}^n (f(x_i)H_i(x) + f'(x_i)K_i(x))$ is of degree $2n+1$

also

$$p_{2n+1}(x_i) = \sum_{i=0}^n (f(x_i)H_i(x_i) + f'(x_i)K_i(x_i))$$

$$p_{2n+1}(x_i) = f(x_i)$$

$\therefore$ solves HPIP

$$K_i(x) = \underbrace{[L_i(x)]^2}_{2n} \underbrace{(x-x_i)}_{+1}$$

• degree of $K_i(x)$ is $2n+1$

$$\begin{aligned} K_i'(x) &= [L_i(x)]^2 + x(2L_i(x))L_i'(x) - x_i(2L_i(x))L_i'(x) \\ &= [L_i(x)]^2 + (x-x_i)(2L_i(x)L_i'(x)) \end{aligned}$$

if $i=k$

$$\begin{aligned} K_i(x_i) &= [L_i(x_i)]^2 (x_i - x_i) \\ &= (1)(0) \\ &= 0 \end{aligned}$$

$$\begin{aligned} K_i'(x_i) &= [L_i(x_i)]^2 + (x_i - x_i)(2L_i(x_i)L_i'(x_i)) \\ &= 1^2 + 0 \\ &= 1 \end{aligned}$$

if $i \neq k$

$$\begin{aligned} K_i(x_k) &= [L_i(x_k)]^2 (x_k - x_i) \\ &= 0 \end{aligned}$$

$$\begin{aligned} K_i'(x_k) &= [L_i(x_k)]^2 + (x_k - x_i)(2L_i(x_k)L_i'(x_k)) \\ &= 0 + (x_k - x_i)(0) \\ &= 0 \end{aligned}$$

$$K_i(x_k) = 0$$

$$K_i'(x_k) = \begin{cases} 1 & i=k \\ 0 & i \neq k \end{cases}$$

Q6. $f(x) = x^5$, $x_0 = 0$ and $x_1 = a$

$$P_3(x) = f(x_0)H_0(x) + f'(x_0)K_0(x) + f(x_1)H_1(x) + f'(x_1)K_1(x)$$

$$f(x_0) = f(0) = 0 \quad f(x_1) = f(a) = a^5$$

$$f'(x_0) = f'(0) = 0 \quad f'(x_1) = f'(a) = 5a^4$$

$$\therefore P_3(x) = f(x_1)H_1(x) + f'(x_1)K_1(x)$$

$$L_1(x) = \frac{x-x_0}{x_1-x_0} = \frac{x}{a}$$

$$L_1'(x) = \frac{1}{a}$$

$$H_1(x) = [L_1(x)]^2 (1 - 2L_1'(x)(x-x_1))$$

$$K_1(x) = [L_1(x)]^2 (x-x_1)$$

$$H_1(x) = [L_1(x)]^2 (1 - 2L_1'(x_1)(x-x_1))$$

$$= \left(\frac{x}{a}\right)^2 \left(1 - \left(\frac{2}{a}\right)(x-a)\right)$$

$$= \frac{x^2}{a^2} \left(1 - \frac{2x}{a} + 2\right)$$

$$= \frac{x^2}{a^2} - \frac{2x^3}{a^3} + \frac{2x^2}{a^2}$$

$$= \frac{3x^2}{a^2} - \frac{2x^3}{a^3}$$

$$K_1(x) = [L_1(x)]^2 (x-x_1)$$

$$= \left(\frac{x}{a}\right)^2 (x-a)$$

$$= \frac{x^3}{a^2} (x-a)$$

$$= \frac{x^3}{a^2} - \frac{x^2}{a}$$

$$P_3(x) = f(x_1)H_1(x) + f'(x_1)K_1(x)$$

$$= (a^5) \left(\frac{3x^2}{a^2} - \frac{2x^3}{a^3} \right) + 5a^4 \left(\frac{x^3}{a^2} - \frac{x^2}{a} \right)$$

$$= 3a^3x^2 - 2a^2x^3 + 5a^2x^3 - 5a^3x^2$$

$$= 3a^2x^3 - 2a^3x^2$$